

Time-Series Momentum in a Ten-ETF Universe

Volatility targeting, regime overlay and crisis performance, 2008 to 2026: one-page summary

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github.com/zack-higham/Time-Series-Momentum-with-Volatility-Targeting-and-Regime-Conditioned-Exposure

Question

Does a multi-asset time-series momentum (TSMOM) strategy built only from liquid ETFs earn a positive risk-adjusted return, which layer of the strategy drives it, does it pay off when equities fall, and is it worth holding beside a conventional portfolio?

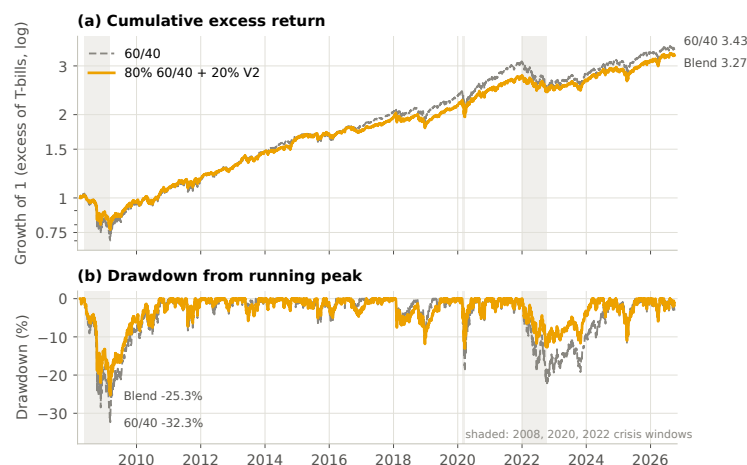
Data and design

Ten ETFs in four asset classes (equities SPY, EFA, EEM; Treasuries IEF, TLT; commodities GLD, DBC; currencies UUP, FXF, FXA), daily total returns in excess of T-bills, 222 months from April 2008 to September 2026, 10 bps per unit traded. Every parameter was committed to git before the first backtest.

- **Signal (V1):** long or short each ETF by the sign of its trailing 12-month excess return.
- **Volatility targeting (V2, primary):** equal ex-ante risk per asset class, scaled to 10% portfolio volatility.
- **Regime overlay (V3):** exposure scaled by a walk-forward Markov-switching model's filtered probability that trend is paying.

Findings

1. **Volatility targeting is the layer that matters.** V2: Sharpe 0.45 (V1: 0.26), maximum drawdown -26.8% , monthly skew turned from -0.50 to $+0.33$; the gain over V1 is not significant (bootstrap $p = 0.19$). Correlation 0.77 with the AQR futures TSMOM factor validates the ETF implementation.
2. **Crisis performance.** V2 gained 9.2%, 12.6% and 30.6% in the three largest equity drawdowns (2008 to 2009, 2020, 2022) while SPY fell 51.8%, 33.8% and 25.3%.
3. **As a diversifier (pre-specified 80/20 blend).** Adding 20% V2 to a 60/40 portfolio cut its maximum drawdown from -32.3% to -25.3% and its loss in all six equity crisis windows (2022: -21.6% to -12.6%), at a lower mean. Sharpe rose from 0.74 to 0.86, not significant ($p = 0.08$).
4. **Regime timing fails out of sample.** The walk-forward overlay lowers the Sharpe ratio to 0.39, and normalised to hold V2's average exposure it still trails V2 (0.40, $p = 0.33$); the same overlay with full-sample smoothed probabilities would have shown a spurious 0.57, an illustration of lookahead bias.
5. **Honest significance.** Deflated for the 26 configurations examined, V2's Sharpe ratio is not significant (DSR 0.68).



60/40 against 80% 60/40 + 20% V2: growth in excess of T-bills and drawdowns (shaded: crisis windows).

Main limitation

One 18.5-year sample with one long bond rally and one sharp rate rise, the episodes behind both V2's crisis gains and the blend's lower drawdowns; ten ETFs give low breadth, and a Sharpe standard error of about 0.23 means no comparison between variants is statistically significant. The ETF implementation also needs leverage (gross 2.15) and shorting, whose costs can remove a quarter of the Sharpe ratio.